How Vaccines Teach the Immune System

The problem vaccines solve

When a new pathogen enters the body for the first time, the immune system has no ready-made defence against it. It must first recognise the invader, then build a targeted response, and that process takes days. During those days the pathogen can multiply freely, which is why a first infection with a dangerous virus can be severe or even fatal. A vaccine gives the immune system a safe preview of the pathogen, so that the slow first response happens in advance, without the illness.

The idea is old. In the eighteenth century, people noticed that milkmaids who had caught the mild disease cowpox seldom caught smallpox. Edward Jenner tested this in 1796 by deliberately exposing a boy to cowpox and later to smallpox; the boy did not fall ill. The word vaccine comes from vacca, the Latin word for cow.

Innate and adaptive immunity

The immune system has two broad layers. The innate layer reacts within minutes to hours and is the same for every invader: inflammation, fever, and cells such as neutrophils and macrophages that engulf anything that looks foreign. It is fast but not specific, and it does not remember what it has seen.

The adaptive layer is slower but precise. Its main players are B cells, which produce antibodies, and T cells, which either help coordinate the response or kill infected cells directly. Each B or T cell carries a receptor that fits one particular molecular shape, called an antigen. The body maintains millions of different receptor shapes, so for almost any antigen there are a few cells that fit it.

Clonal selection, affinity maturation and memory

When an antigen-presenting cell, usually a dendritic cell, carries a fragment of the pathogen to a lymph node, the rare B and T cells whose receptors match that fragment are activated and begin to divide rapidly, a process called clonal selection and expansion. Within roughly seven to fourteen days a single matching cell can become thousands of identical copies. Inside structures called germinal centres, B cells also undergo somatic hypermutation: their antibody genes mutate at a high rate, and only the variants that bind the antigen more tightly receive survival signals from follicular helper T cells. This repeated mutation and selection, known as affinity maturation, means the antibodies produced late in a response can bind the antigen ten to a hundred times more strongly than the first ones. Activated B cells differentiate either into plasma cells, which secrete up to two thousand antibody molecules per second but live only days to weeks in most cases, or into long-lived memory B cells. A similar split occurs among T cells, producing effector cells and memory T cells. After the infection is cleared, most effector cells die off through apoptosis, but the memory cells persist, sometimes for decades, and a second encounter with the same antigen triggers a secondary response that is faster, larger

and of higher affinity than the primary one, often clearing the pathogen before any symptoms appear.

Types of vaccine

Every vaccine presents antigens without causing the disease, but they do it in different ways. Live attenuated vaccines use a weakened form of the pathogen, as in the measles vaccine. Inactivated vaccines use a killed pathogen. Subunit vaccines contain only selected proteins. mRNA vaccines deliver instructions that let the body's own cells make one viral protein for a short time.

Many vaccines need more than one dose. The first dose triggers the primary response; a booster weeks or months later triggers a secondary response that raises antibody levels and produces more memory cells.

Herd immunity

When enough people in a community are immune, a pathogen cannot find enough susceptible hosts to keep spreading, so even people who cannot be vaccinated, such as newborns, are protected. The share of the population that must be immune depends on how contagious the disease is. For measles, one of the most contagious diseases known, it is about 95 percent.